

Groundwater storage: modeled outcome levels across 28 management strategies and five hydroclimates

A results brief for a general reader — prototype (v0_1), built by the brief builder from the locked 2026-08-29 API snapshot — reporting modeled outcome levels for one sector of the water system under every management strategy in the ensemble, read across the five hydroclimates, each difference from the baseline split into the two effects that produce it. The build stamp, the run identifiers and the verification status are in references.

TL;DR. Compared with the baseline — current operations under the historical hydroclimate — modeled outcome levels for groundwater storage differ from the baseline in all 28 management strategies under at least one hydroclimate. Reading each strategy under each hydroclimate as one result, 140 in all, the hydroclimate effect is the larger of the two in 72, the operations effect in 66, and neither is larger in the remaining 2. Of the 42 locations behind this sector, the number a strategy changes runs from 0 — Current operations, at the hydroclimate where it reaches fewest — to 40, at Maintain Delta outflows (45% unimpaired flow target) where it reaches most. Groundwater pumping limits via reduced crop acreage in the Central Valley is the clearest case of the two effects running opposite ways: under high climate stress the hydroclimate contributes +0.3 and the strategy −0.4, leaving a difference from the baseline of −0.1.

What this is

This is a *results brief*: it reports the modeled outcome levels, and nothing else, for one sector of the water system — groundwater storage — under every management strategy in the ensemble, each simulated once under each of five hydroclimates. It transposes the grid the strategy results briefs report: one sector, every strategy, rather than one strategy, every sector. The selection rule is that there is none: this brief reports every management strategy these model runs carry an outcome level for, across all five modeled hydroclimates, and no filter is applied to either. All 28 management strategies are reported under every hydroclimate. A scenario is a strategy paired with one hydroclimate; the five hydroclimates are a sample of stress levels, not a forecast of which will occur. For how the five were constructed, see the hydroclimates reference brief.

What the model shows

Across the 28 strategies these runs report, the modeled level value under the baseline — current operations under the historical hydroclimate — is 2.7, the same value for every row. Read across the five hydroclimates, the values this brief reports range from 2.1 to 3.5. Of the 140 results — one strategy under one hydroclimate — 56 sit in a different level band from the baseline and 84 sit in the same band; in all 28 strategies the five values do not all sit in one band.

The figure below draws every one of those results beside the baseline, with the two effects that produce the difference shown as separate marks, so a reader can see how much of each difference the hydroclimate carries and how much the strategy does. The section that follows says what the two effects are before the figure draws them; what the level values themselves mean is set out in how to read the values.

The baseline, and the two effects

The baseline is current operations under the historical hydroclimate. It is the one reference this brief asks a reader to learn, and it is the same value in every column, because it does not depend on which hydroclimate a scenario pairs with.

Any difference between a scenario and the baseline comes from two places, and they add exactly:

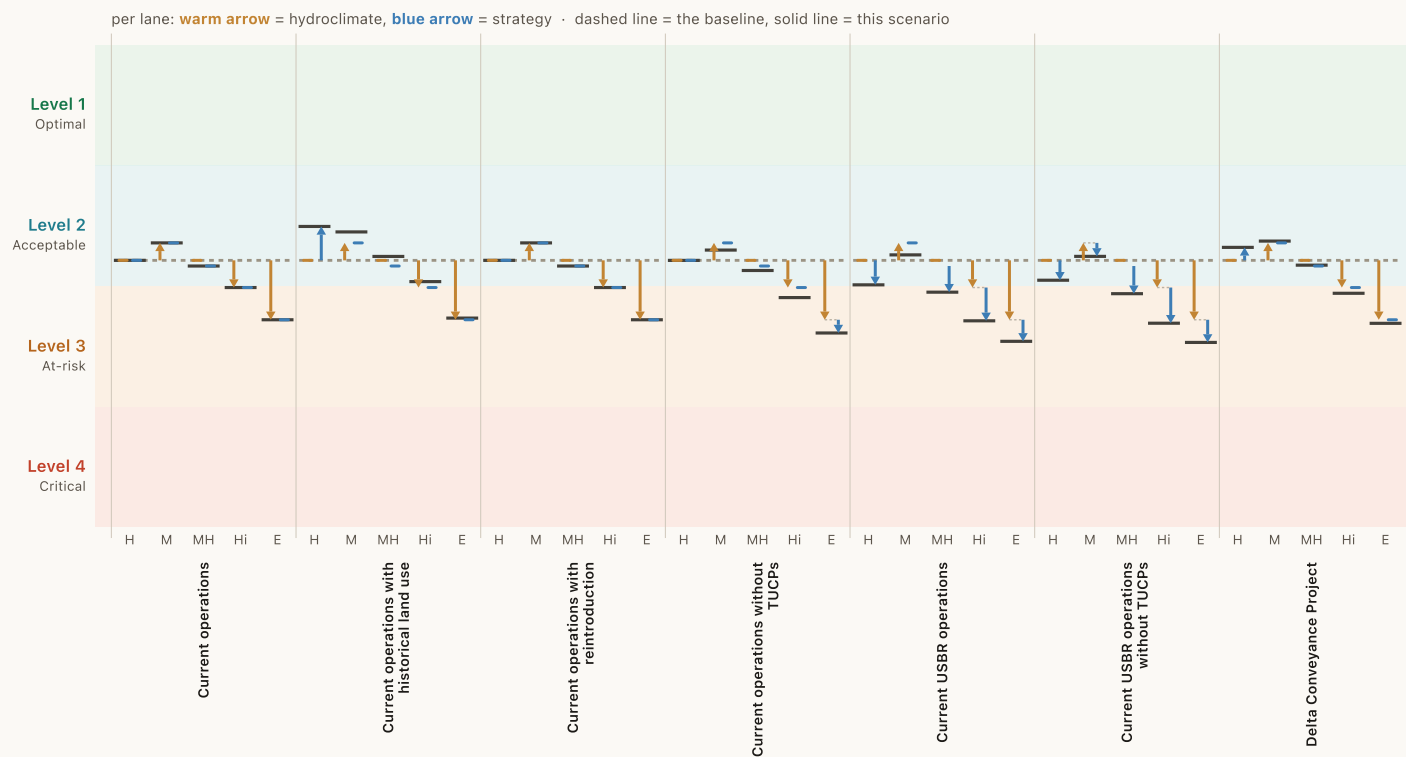
The hydroclimate effect is current operations under that hydroclimate minus the baseline — the same operations, a different hydroclimate. **The operations effect** is that strategy minus current operations *under that same hydroclimate* — the only comparison that holds the hydroclimate fixed, and therefore the only one that isolates what the strategy does. Together they equal the difference from the baseline.

Outcome levels are reported as a value on a continuous axis from Level 1 to Level 4.99, with the four level bands as shaded zones; the whole number below any value is its band. A value that lands exactly on a whole number takes that band: 4.0 is Level 4, at the start of its zone rather than the top of Level 3. The level scale is ordered by definition, Level 1 through Level 4; a band is a category and a position on the axis is a modeled magnitude, neither is a judgment. The scale, the two thresholds and the two fields a level value comes in are set out in how to read the values.

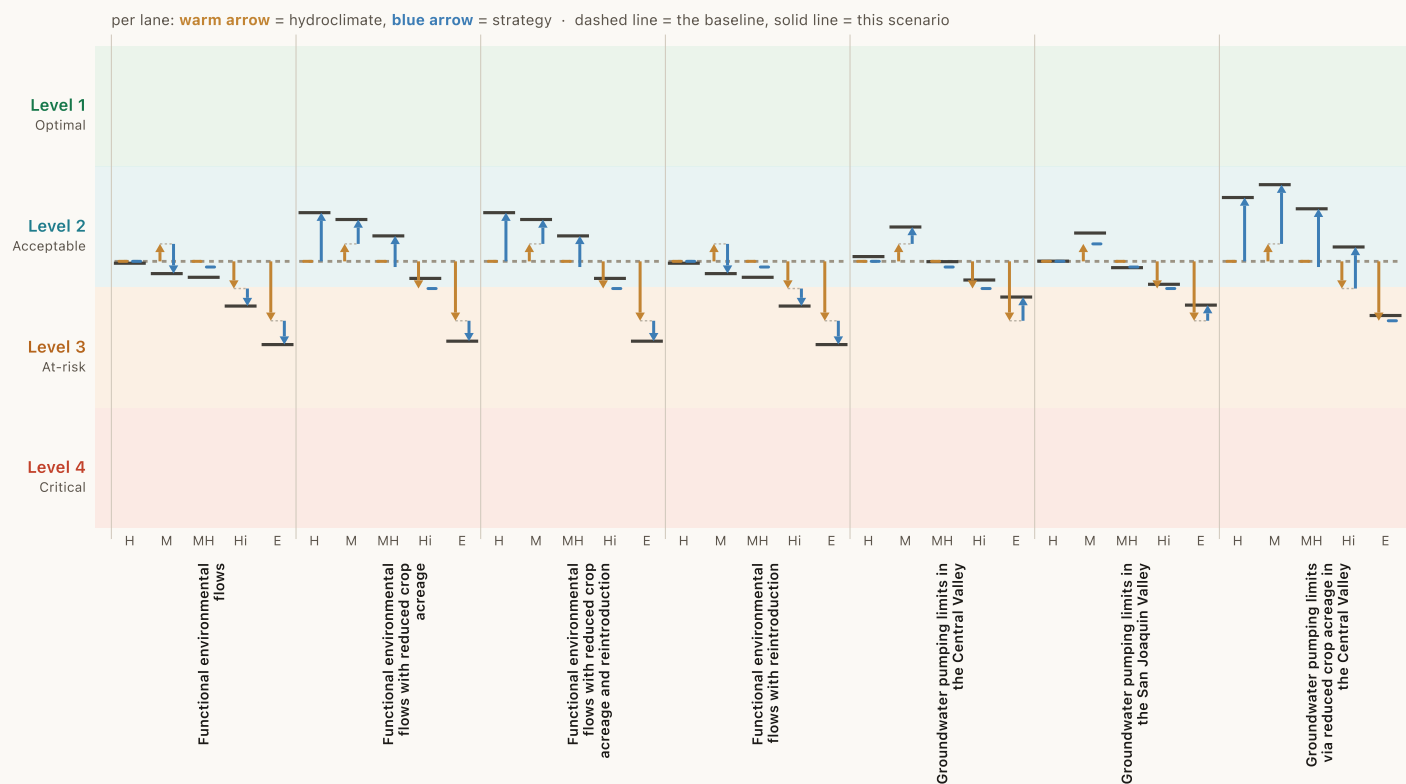
Groundwater storage's level value is the unweighted mean of the level values at its 42 locations, and every row on this page is measured over the same 42: the rows differ in which strategy produced the values, never in what was measured. Every location counts equally, regardless of the people, acres or acre-feet it represents. This page reports sectors, not individual locations; where a place brief has been published for one of a sector's locations, that brief reports the location's own values across the same scenarios.

All 28 strategies, all five hydroclimates

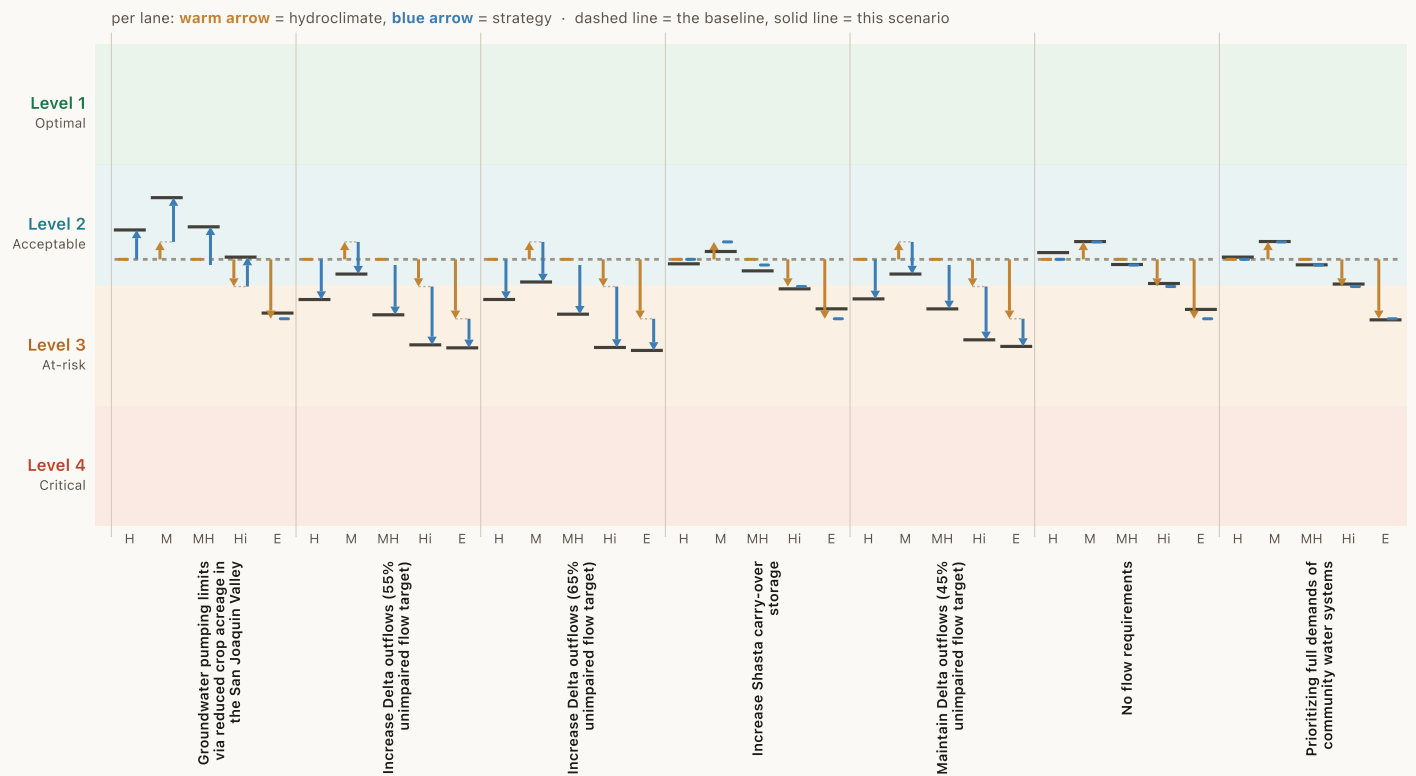
Panel 1 of 4 — strategies 1 to 7 of 28



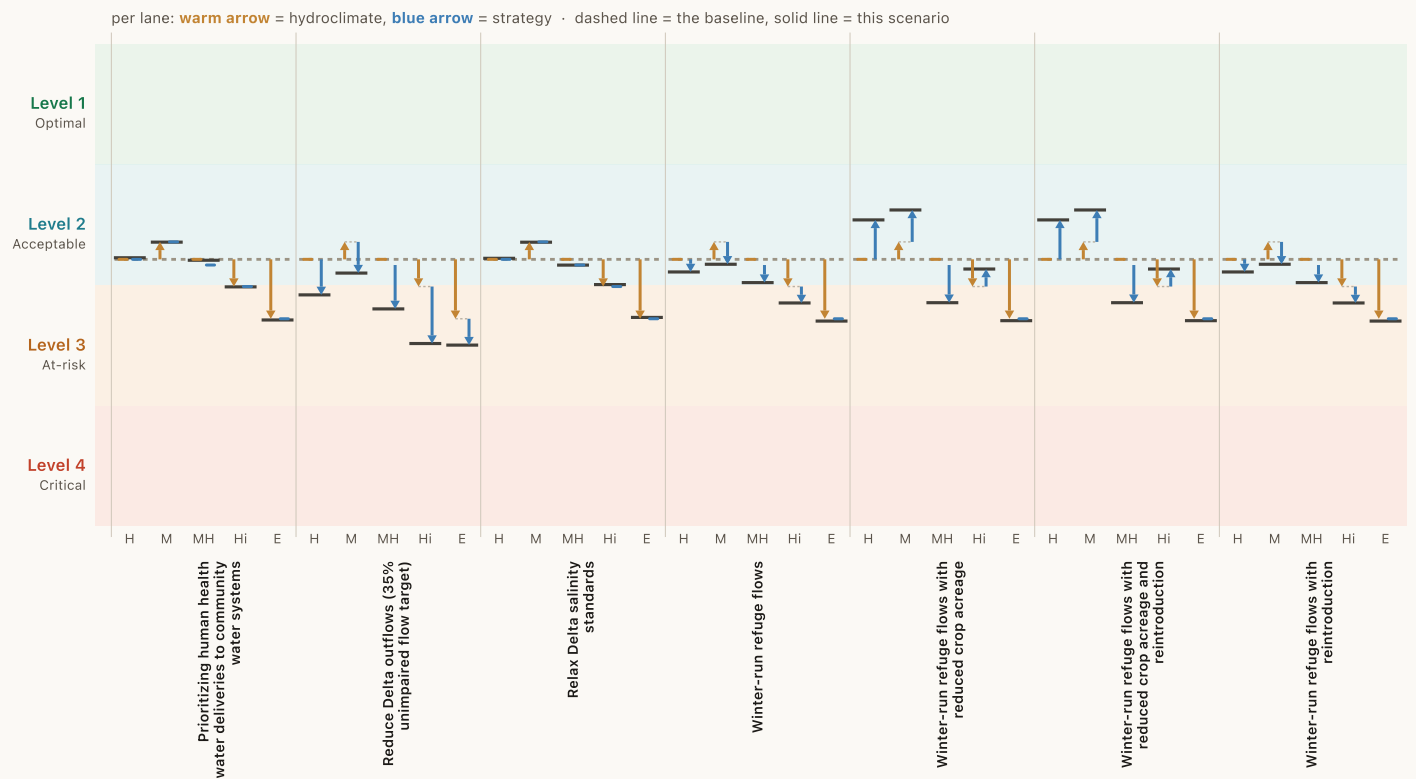
Panel 2 of 4 — strategies 8 to 14 of 28



Panel 3 of 4 — strategies 15 to 21 of 28



Panel 4 of 4 — strategies 22 to 28 of 28



The 28 strategies are shown in four stacked panels that share one level axis. The split is positional and carries no meaning; the panels are read as one figure continued.

Figure 1. Modeled level values for groundwater storage and the two effects that produce their difference from the baseline, 28 management strategies $\times$ five hydroclimates. Lanes within each strategy run left to right: Historical (H), Moderate (M), Moderate-high (MH), High (Hi), Extreme (E) climate stress. Vertical axis: continuous level value, Level 1 at top, level bands as shaded zones. Dashed level: the baseline. Solid level: this scenario. A short flat tick in place of an arrow is a change smaller than 0.1 on the level scale. Every value is in the table that follows, and the scale is set out in how to read the values.

The same 140 results as numbers

Each cell reads top to bottom: the hydroclimate effect, the operations effect, and the resulting value with its band. The two effects add to the difference between the scenario and the baseline. Outlined cells are those where the two run opposite ways. Each hydroclimate column carries its own address for citation, and its own slice of the companion data file.

Strategy	The baseline	Historical	Moderate	Moderate-high	High	Extreme	Operations effect across the five
Current operations	L2 2.7	0.0 0.0 L2 2.7	-0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L3 3.0	+0.5 0.0 L3 3.2	no change at the 0.1 threshold 0.0
Current operations with historical land use	L2 2.7	0.0 -0.2 L2 2.5	-0.1 0.0 L2 2.5	0.0 0.0 L2 2.7	+0.3 0.0 L2 2.9	+0.5 0.0 L3 3.2	one direction -0.2 to 0.0
Current operations with reintroduction	L2 2.7	0.0 0.0 L2 2.7	-0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L3 3.0	+0.5 0.0 L3 3.2	no change at the 0.1 threshold 0.0
Current operations without TUCPs	L2 2.7	0.0 0.0 L2 2.7	-0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L3 3.0	+0.5 +0.1 L3 3.3	one direction 0.0 to +0.1
Current USBR operations	L2 2.7	0.0 +0.2 L2 2.9	-0.1 0.0 L2 2.7	0.0 +0.2 L3 3.0	+0.3 +0.2 L3 3.2	+0.5 +0.2 L3 3.4	one direction 0.0 to +0.2
Current USBR operations without TUCPs	L2 2.7	0.0 +0.2 L2 2.9	-0.1 +0.1 L2 2.7	0.0 +0.2 L3 3.0	+0.3 +0.3 L3 3.3	+0.5 +0.2 L3 3.4	one direction +0.1 to +0.3
Delta Conveyance Project	L2 2.7	0.0 -0.1 L2 2.6	-0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L3 3.0	+0.5 0.0 L3 3.3	one direction -0.1 to 0.0
Functional environmental flows	L2 2.7	0.0 0.0 L2 2.8	-0.1 +0.2 L2 2.8	0.0 0.0 L2 2.9	+0.3 +0.1 L3 3.1	+0.5 +0.2 L3 3.4	one direction 0.0 to +0.2
Functional environmental flows with reduced crop acreage	L2 2.7	0.0 -0.4 L2 2.3	-0.1 -0.2 L2 2.4	0.0 -0.3 L2 2.5	+0.3 0.0 L2 2.9	+0.5 +0.2 L3 3.4	crosses zero -0.4 to +0.2
Functional environmental flows with reduced crop acreage and reintroduction	L2 2.7	0.0 -0.4 L2 2.3	-0.1 -0.2 L2 2.4	0.0 -0.3 L2 2.5	+0.3 0.0 L2 2.9	+0.5 +0.2 L3 3.4	crosses zero -0.4 to +0.2
Functional environmental flows with reintroduction	L2 2.7	0.0 0.0 L2 2.8	-0.1 +0.2 L2 2.8	0.0 0.0 L2 2.9	+0.3 +0.1 L3 3.1	+0.5 +0.2 L3 3.4	one direction 0.0 to +0.2

Strategy	The baseline	Historical	Moderate	Moderate-high	High	Extreme	Operations effect across the five
Groundwater pumping limits in the Central Valley	 2.7	 0.0 0.0	 -0.1 -0.1	 0.0 0.0	 +0.3 0.0	 +0.5 -0.2 3.0	one direction -0.2 to 0.0
Groundwater pumping limits in the San Joaquin Valley	 2.7	 0.0 0.0	 -0.1 0.0	 0.0 0.0	 +0.3 0.0	 +0.5 -0.1 3.1	one direction -0.1 to 0.0
Groundwater pumping limits via reduced crop acreage in the Central Valley	 2.7	 0.0 -0.5	 -0.1 -0.5	 0.0 -0.5	 +0.3 -0.4	 +0.5 0.0 3.2	one direction -0.5 to 0.0
Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley	 2.7	 0.0 -0.2	 -0.1 -0.4	 0.0 -0.3	 +0.3 -0.3	 +0.5 0.0 3.2	one direction -0.4 to 0.0
Increase Delta outflows (55% unimpaired flow target)	 2.7	 0.0 +0.4	 -0.1 +0.3	 0.0 +0.4	 +0.3 +0.4	 +0.5 +0.3 3.5	one direction +0.3 to +0.4
Increase Delta outflows (65% unimpaired flow target)	 2.7	 0.0 +0.4	 -0.1 +0.3	 0.0 +0.4	 +0.3 +0.5	 +0.5 +0.3 3.5	one direction +0.3 to +0.5
Increase Shasta carry-over storage	 2.7	 0.0 0.0	 -0.1 0.0	 0.0 0.0	 +0.3 0.0	 +0.5 0.0 3.1	no change at the 0.1 threshold 0.0
Maintain Delta outflows (45% unimpaired flow target)	 2.7	 0.0 +0.4	 -0.1 +0.3	 0.0 +0.3	 +0.3 +0.4	 +0.5 +0.3 3.5	one direction +0.3 to +0.4
No flow requirements	 2.7	 0.0 0.0	 -0.1 0.0	 0.0 0.0	 +0.3 0.0	 +0.5 0.0 3.1	no change at the 0.1 threshold 0.0
Prioritizing full demands of community water systems	 2.7	 0.0 0.0	 -0.1 0.0	 0.0 0.0	 +0.3 0.0	 +0.5 0.0 3.2	no change at the 0.1 threshold 0.0
Prioritizing human health water deliveries to community water systems	 2.7	 0.0 0.0	 -0.1 0.0	 0.0 0.0	 +0.3 0.0	 +0.5 0.0 3.2	no change at the 0.1 threshold 0.0

Strategy	The baseline	Historical	Moderate	Moderate-high	High	Extreme	Operations effect across the five
Reduce Delta outflows (35% unimpaired flow target)	L2 2.7	0.0 +0.3 L3 3.0	-0.1 +0.2 L2 2.8	0.0 +0.3 L3 3.1	+0.3 +0.4 L3 3.4	+0.5 +0.2 L3 3.4	one direction +0.2 to +0.4
Relax Delta salinity standards	L2 2.7	0.0 0.0 L2 2.7	-0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L2 2.9	+0.5 0.0 L3 3.2	no change at the 0.1 threshold 0.0
Winter-run refuge flows	L2 2.7	0.0 +0.1 L2 2.8	-0.1 +0.2 L2 2.8	0.0 +0.1 L2 2.9	+0.3 +0.1 L3 3.1	+0.5 0.0 L3 3.2	one direction 0.0 to +0.2
Winter-run refuge flows with reduced crop acreage	L2 2.7	0.0 -0.3 L2 2.4	-0.1 -0.3 L2 2.3	0.0 +0.3 L3 3.1	+0.3 -0.2 L2 2.8	+0.5 0.0 L3 3.2	crosses zero -0.3 to +0.3
Winter-run refuge flows with reduced crop acreage and reintroduction	L2 2.7	0.0 -0.3 L2 2.4	-0.1 -0.3 L2 2.3	0.0 +0.3 L3 3.1	+0.3 -0.2 L2 2.8	+0.5 0.0 L3 3.2	crosses zero -0.3 to +0.3
Winter-run refuge flows with reintroduction	L2 2.7	0.0 +0.1 L2 2.8	-0.1 +0.2 L2 2.8	0.0 +0.1 L2 2.9	+0.3 +0.1 L3 3.1	+0.5 0.0 L3 3.2	one direction 0.0 to +0.2

How many places each strategy reaches

A strategy's rules apply in some places and not others, and this sector's value is the unweighted mean of its locations, so a real change at a few of them reads as a small sector-level difference. A location is counted where its value differs from current operations by at least 0.1 on the level scale. Of the 42 locations behind this sector, the number a strategy changes runs from 0 — Current operations, at the hydroclimate where it reaches fewest — to 40, at Maintain Delta outflows (45% unimpaired flow target) where it reaches most. Two of them differ at no location under any hydroclimate. The per-strategy counts, split by hydroclimate and by direction, and the median difference behind each, are in the companion data file at [#/rows/*/reach](#).

Read across the five hydroclimates together, 2,919 of the 5,880 location-and-hydroclimate results reported here differ from current operations: 1,345 toward Level 1, 1,574 toward Level 4. The median difference over the locations counted runs from 0.1 on the level scale at Relax Delta salinity standards to 1.1 at Increase Delta outflows (55% unimpaired flow target). This page reports sectors, not individual locations; where a place brief has been published for one of a sector's locations, that brief reports the location's own values across the same scenarios.

Related briefs

What happens in another sector? A sector brief reports one sector of the water system, every strategy, the same five hydroclimates:

Agricultural revenue

Community surface water

Delta estuary ecology

Environmental flows

Freshwater for Delta exports

Freshwater for in-Delta uses

Groundwater storage — *you are here*

Reservoir storage

Winter-run salmon

What happens across all nine sectors under one strategy? This brief holds the sector fixed and varies the strategy. The transpose does the reverse, reporting every sector under one strategy: one brief per strategy, and one more that holds the strategy at current operations:

Current operations

Current operations with historical land use

Current operations with reintroduction

Current operations without TUCPs

Current USBR operations

Current USBR operations without TUCPs

Delta Conveyance Project

Functional environmental flows

Functional environmental flows with reduced crop acreage

Functional environmental flows with reduced crop acreage and reintroduction

Functional environmental flows with reintroduction

Groundwater pumping limits in the Central Valley

Groundwater pumping limits in the San Joaquin Valley

Groundwater pumping limits via reduced crop acreage in the Central Valley

Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley

Increase Delta outflows (55% unimpaired flow target)

Increase Delta outflows (65% unimpaired flow target)

Increase Shasta carry-over storage

Maintain Delta outflows (45% unimpaired flow target)

No flow requirements

Prioritizing full demands of community water systems

Prioritizing human health water deliveries to community water systems

Reduce Delta outflows (35% unimpaired flow target)

Relax Delta salinity standards

The water system under current operations

Winter-run refuge flows

Winter-run refuge flows with reduced crop acreage

Winter-run refuge flows with reduced crop acreage and reintroduction

Winter-run refuge flows with reintroduction

How were the five hydroclimates constructed? The hydroclimates reference brief defines them. It reports no results of its own; every brief above is read against the same five.

What this doesn't say

This brief reports modeled results from the COEQWAL ensemble. It does not predict what will happen, evaluate whether those results are acceptable, or recommend any course of action. The outcome-level framework provides definitional categories; the meaning of those categories for any individual reader, community, or decision is not specified by COEQWAL.

In particular, for this brief: the five hydroclimates are a sample of stress levels, carry no probabilities, and are not a timeline, so no average over them is reported here. Splitting a difference into two effects is arithmetic on modeled values, not a causal claim about the water system, and the two are separable only because each is defined against its own reference. A step or a spread on the level axis is a modeled magnitude, not a benefit or a cost. A sector's value is the unweighted mean of its locations, so locations count equally regardless of the people, acres, or acre-feet each represents, and a location already at either end of the scale cannot move further along it. Locations are modeled nodes, not measured conditions at a place.

Extreme climate stress holds the single highest of a strategy's five values in all 28 strategies reported here.

References, run identifiers and the build stamp

Model runs (current operations): Historical `s0020` ; Moderate `s0134` ; Moderate-high `s0047` ; High `s0056` ; Extreme `s0108` . Each row reads its own five runs beside these; they are in the companion data file at `#/rows/*/by_hydroclimate/*/run_ids` . The baseline is current operations under the historical hydroclimate, `s0020` .

Build stamp. Snapshot `2026-08-29` ; builder version `0.1.0` ; built 2026-08-29T16:19:04Z; snapshot data read 2026-08-29; verification: pass. The verification status is written by the independent check (`briefbuilder.verifier`), which recomputes every value in the companion data file from the per-location data and reads every arrow on this page, after both files are written. It is not stamped by whatever produced them.

Companion data file: `sector_groundwater-storage_schema1.json` , holding every value in this brief plus the per-location counts, the spread measures, the derivation of every number in the prose and the figure's declared attribute contract. Per-hydroclimate slices: Historical `#/rows/*/by_hydroclimate/BL` ; Moderate `#/rows/*/by_hydroclimate/H4` ; Moderate-high `#/rows/*/by_hydroclimate/H1` ; High `#/rows/*/by_hydroclimate/H2` ; Extreme `#/rows/*/by_hydroclimate/H3` . Built from the locked 2026-08-29 API snapshot (api.coeqwal.org v2.1.1 (locked dataset snapshot)). Decomposition per `Decomposition_Math_Quickref` (coeqwal-viz): the hydroclimate effect is current operations under that hydroclimate minus current operations under the historical hydroclimate; the operations effect is this strategy minus current operations under that same hydroclimate. Axis convention [1, 4.99] per `Ordinal_vs_continuous_tiers.md` ; the visible band is `floor(value)` . A difference of 0.1 on the level scale is required before a difference is reported as movement, before a location is counted as differing from its comparison, and before an effect prints as anything other than 0.0. Color semantics (warm = hydroclimate, blue = operations) follow the ops-vs-climate stepper. Strategy identity is its descriptive name; run IDs appear only in this note and the data file.

How to read the values

The level scale runs from 1 to 4.99. Level 1 is the favorable end of the scale: the outcome-level framework names Level 1 Optimal and Level 4 Critical, so a smaller value sits nearer Optimal and a larger value nearer Critical. The API serves this number in a field named `weighted_score`; that name describes how the number is computed, and does not mean that a larger number is a more favorable result.

Two numbers describe each result. The integer `tier_level` is the level label. `tier_continuous` is the position within the band of that label, and always falls inside it: `tier_continuous` is at least `tier_level` and less than `tier_level` plus one. Labels, counts and prose use `tier_level`; figure positions and statistics use `tier_continuous`.

Level 1 is Optimal, Level 2 is Acceptable, Level 3 is At-risk, and Level 4 is Critical. These are definitional category names supplied by the outcome-level framework; what a category means for any reader, community, or decision is not specified by COEQWAL.

A sector value is the unweighted mean over the locations in that sector. Every location counts equally, regardless of the people, acres, or acre-feet it represents.

Any difference between two values under different hydroclimates contains both a hydroclimate effect and an operations effect; the two are separable only through the decomposition, because each is defined against its own reference. A strategy is compared with current operations under the same hydroclimate. Comparing one strategy with itself across hydroclimates is a different quantity: the hydroclimate effect plus the change in the operations effect.

One threshold governs every claim and every count here. A difference smaller than 0.1 on the level scale is not reported as movement, and a location is not counted as differing from its comparison. An outcome level is a summary built so that results in nine sectors with no shared unit can be compared, and 0.1 on the level scale is the finest reading it carries; a smaller difference is left unstated rather than described.

The five hydroclimates are a purposive sample of climate stress levels. They are not a sequence, a trend, or a forecast, they carry no probabilities, and no mean is taken over them.

A run ID identifies one modeling run of one strategy under one hydroclimate. It is an internal identifier, used for citation and for tracing a value back to the data platform, not a name for the strategy; the public identity of a strategy is its descriptive name.

Level values are reported to one decimal place throughout this brief — in the prose, in the figure and in the table. The level scale does not carry meaning finer than that. Each effect shown is the difference between the floored values, except that an effect smaller than 0.1 on the level scale, which is the threshold this brief reports movement at, is shown as 0.0, so a row containing one does not add up. Values are floored rather than rounded, so a printed value stays inside the band its own label names; full-precision values are in the machine-readable sidecar this brief is built from.